ModelIt!

Student Activity Pack

Newton's Laws: Why Things Move the Way They Do

The Three Rules That Govern Every Motion in the Universe

Grade 07L2 | NGSS MS-PS2-1, MS-PS2-2 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why is it harder to push a full shopping cart than an empty one?

2 What makes a moving ball eventually stop on a flat surface?

3 Draw what you think happens to the speed of an object when you push it harder.

4 If you push two objects with the same force, which moves faster: the heavy one or the light one? Why?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Applied Force | Object Mass | Acceleration | Friction Force | Net Force

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Light + Strong Force

High applied force, low mass, moderate friction

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Heavy + Same Force

Same applied force, much higher mass, same friction

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: High Friction

Moderate force, moderate mass, high friction

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Air Resistance — The drag force exerted by air on a moving object, increasing with speed and surface area

Gravitational Force — The downward pull of gravity on all objects with mass, equal to mass times gravitational acceleration

Momentum — The quantity of motion an object has, calculated as mass times velocity, which is conserved in collisions

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does a bowling ball keep rolling while a tennis ball stops quickly, and why does pushing a shopping cart feel different when it's full vs. empty? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that F = ma? What evidence from the simulation proves this relationship?**

2. **Why does a hockey puck slide so much farther on ice than a ball rolls on grass?**

3. **What would motion look like in a world with zero friction?**

4. **How do Newton's Laws explain why seatbelts save lives in car crashes?**

5. **If two objects of different masses fall from the same height, why do they hit the ground at the same time (ignoring air resistance)?**

8 STEM Challenge: Design a Vehicle Safety Test

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YOUR ENGINEERING MISSION

Design and conduct a vehicle safety test demonstrating how mass, force, and friction affect collision outcomes, applying Newton's Laws to real-world engineering.

THE SCENARIO

An automotive safety company needs to test how different vehicle masses and speeds affect collision impacts. Your team will design a scaled-down crash test using toy cars, ramps, and barriers to demonstrate Newton's Laws in action.

Engineering Design Thinking

1. WHAT VARIABLES WILL YOU TEST IN YOUR CRASH TEST (MASS, SPEED, SURFACE TYPE)?
2. HOW WILL YOU MEASURE THE FORCE OF IMPACT IN YOUR SCALED MODEL?
3. WHAT SAFETY FEATURES COULD REDUCE IMPACT FORCE BASED ON NEWTON'S LAWS?
4. HOW WILL YOU RECORD AND PRESENT YOUR DATA AS EVIDENCE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

Explain your design using scientific evidence from the simulation:

Real Career Connection

Mechanical Engineers apply Newton's Laws to design machines, vehicles, and structures. They analyze forces, motion, and materials to create safe, efficient products, earning $75,000-$130,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 Two identical forces are applied to two different objects. Object A has a mass of 2 kg and Object B has a mass of 8 kg. Using the model's demonstration of F = ma, which statement correctly compares their accelerations?

- **A.** Both objects accelerate at the same rate because the same force was applied
- **B.** Object A accelerates four times faster than Object B because acceleration is inversely proportional to mass
- **C.** Object B accelerates faster because heavier objects have more momentum
- **D.** Neither object accelerates because friction cancels out the applied force

2 In the model, increasing Friction Force causes Net Force to decrease even when Applied Force stays constant. A student observes that a hockey puck slides much farther on ice than a ball rolls on grass with the same initial push. Which model relationship explains this?

- **A.** Applied Force to Net Force is negative on grass but positive on ice
- **B.** Friction Force has a negative relationship with Net Force, and ice produces much less friction than grass, so more net force remains to sustain motion
- **C.** Object Mass is lower on ice because the surface is smoother
- **D.** Acceleration is not affected by surface type according to the model

3 A loaded truck traveling at 60 mph takes much longer to stop than an empty car traveling at the same speed, even when both apply maximum brakes. Which combination of Newton's Laws best explains this observation?

- **A.** Newton's First Law only: the truck has more weight pressing on the road
- **B.** Newton's First Law (greater mass means greater inertia resisting the change in motion) combined with Newton's Second Law (same braking force applied to greater mass produces less deceleration)
- **C.** Newton's Third Law only: the road pushes back harder on the truck
- **D.** None of Newton's Laws apply because braking is a separate phenomenon from motion

4 In the model, a student sets Applied Force to 10 N and Object Mass to 5 kg, but observes that the actual Acceleration is less than the expected 2 m/s squared. Which model component accounts for the difference between expected and observed acceleration?

- **A.** Net Force, because gravity reduces the applied force on horizontal surfaces
- **B.** Friction Force, which opposes motion and reduces the net force available for acceleration below the full applied force
- **C.** Object Mass, which must have increased automatically during the simulation
- **D.** Acceleration, which has a maximum limit regardless of force and mass values

5 **A student pushes a friend on a skateboard. Newton's Third Law states that the friend pushes back on the student with equal force. The student asks: 'If the forces are equal, why does anyone move?' Which response correctly resolves this apparent paradox?**

- **A.** The forces are not actually equal; the pusher always exerts more force
- **B.** The action and reaction forces act on different objects, so they cannot cancel each other out. Each object accelerates based on the net forces acting on IT
- **C.** Friction makes the reaction force weaker than the action force
- **D.** Newton's Third Law only applies to stationary objects, not moving ones